References

AI@Meta. 2024. Llama 3 model card. *https://github.com/meta-llama/llama3/blob/main*.

Saleh Ashkboos, Maximilian L Croci, Marcelo Gennari do Nascimento, Torsten Hoefler, and James Hensman. 2024. Slicegpt: Compress large language models by deleting rows and columns. *arXiv preprint arXiv:2401.15024*.

Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E Hinton. 2016. Layer normalization. *arXiv preprint arXiv:1607.06450*.

Yonatan Bisk, Rowan Zellers, Jianfeng Gao, Yejin Choi, et al. 2020. Piqa: Reasoning about physical commonsense in natural language. In *Proceedings of the AAAI conference on artificial intelligence*, volume 34, pages 7432–7439.

Tianle Cai, Yuhong Li, Zhengyang Geng, Hongwu Peng, Jason D Lee, Deming Chen, and Tri Dao. 2024. Medusa: Simple llm inference acceleration framework with multiple decoding heads. *arXiv preprint arXiv:2401.10774*.

Zouying Cao, Yifei Yang, and Hai Zhao. 2024. Head-wise shareable attention for large language models. *arXiv preprint arXiv:2402.11819*.

Yanxi Chen, Xuchen Pan, Yaliang Li, Bolin Ding, and Jingren Zhou. 2023. Ee-llm: Large-scale training and inference of early-exit large language models with 3d parallelism. *arXiv preprint arXiv:2312.04916*.

Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. 2018. Think you have solved question answering? try arc, the ai2 reasoning challenge. *arXiv preprint arXiv:1803.05457*.

Mostafa Dehghani, Stephan Gouws, Oriol Vinyals, Jakob Uszkoreit, and Łukasz Kaiser. 2018. Universal transformers. *arXiv preprint arXiv:1807.03819*.

Emily L Denton, Wojciech Zaremba, Joan Bruna, Yann LeCun, and Rob Fergus. 2014. Exploiting linear structure within convolutional networks for efficient evaluation. *Advances in neural information processing systems*, 27.

Mostafa Elhoushi, Akshat Shrivastava, Diana Liskovich, Basil Hosmer, Bram Wasti, Liangzhen Lai, Anas Mahmoud, Bilge Acun, Saurabh Agarwal, Ahmed Roman, et al. 2024. Layer skip: Enabling early exit inference and self-speculative decoding. *arXiv preprint arXiv:2404.16710*.

Leo Gao, Stella Biderman, Sid Black, Laurence Golding, Travis Hoppe, Charles Foster, Jason Phang, Horace He, Anish Thite, Noa Nabeshima, et al. 2020. The Pile: An 800GB dataset of diverse text for language modeling. *arXiv preprint arXiv:2101.00027*.

Leo Gao, Jonathan Tow, Stella Biderman, Sid Black, Anthony DiPofi, Charles Foster, Laurence Golding, Jeffrey Hsu, Kyle McDonell, Niklas Muennighoff, et al. 2021. A framework for few-shot language model evaluation. *Version v0. 0.1. Sept*, page 8.

Muhammad Usman Hadi, Rizwan Qureshi, Abbas Shah, Muhammad Irfan, Anas Zafar, Muhammad Bilal Shaikh, Naveed Akhtar, Jia Wu, Seyedali Mirjalili, et al. 2023. Large language models: a comprehensive survey of its applications, challenges, limitations, and future prospects. *Authorea Preprints*.

Junxian He, Chunting Zhou, Xuezhe Ma, Taylor Berg-Kirkpatrick, and Graham Neubig. 2021. Towards a unified view of parameter-efficient transfer learning. *arXiv preprint arXiv:2110.04366*.

Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin De Laroussilhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly. 2019. Parameter-efficient transfer learning for nlp. In *International Conference on Machine Learning*, pages 2790–2799. PMLR.

Yen-Chang Hsu, Ting Hua, Sungen Chang, Qian Lou, Yilin Shen, and Hongxia Jin. 2022. Language model compression with weighted low-rank factorization. *arXiv preprint arXiv:2207.00112*.

Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2021. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*.

Jean Kaddour. 2023. The minipile challenge for data-efficient language models. *arXiv preprint arXiv:2304.08442*.

Brian Lester, Rami Al-Rfou, and Noah Constant. 2021. The power of scale for parameter-efficient prompt tuning. *arXiv preprint arXiv:2104.08691*.

Yaniv Leviathan, Matan Kalman, and Yossi Matias. 2023. Fast inference from transformers via speculative decoding. In *International Conference on Machine Learning*, pages 19274–19286. PMLR.

Shih-Yang Liu, Chien-Yi Wang, Hongxu Yin, Pavlo Molchanov, Yu-Chiang Frank Wang, Kwang-Ting Cheng, and Min-Hung Chen. 2024a. Dora: Weight-decomposed low-rank adaptation. *arXiv preprint arXiv:2402.09353*.

Xiao Liu, Kaixuan Ji, Yicheng Fu, Weng Lam Tam, Zhengxiao Du, Zhilin Yang, and Jie Tang. 2021. P-tuning v2: Prompt tuning can be comparable to fine-tuning universally across scales and tasks. *arXiv preprint arXiv:2110.07602*.

Zechun Liu, Changsheng Zhao, Forrest Iandola, Chen Lai, Yuandong Tian, Igor Fedorov, Yunyang Xiong, Ernie Chang, Yangyang Shi, Raghuraman Krishnamoorthi, et al. 2024b. Mobilellm: Optimizing sub-billion parameter language models for on-device use cases. *arXiv preprint arXiv:2402.14905*.

Xinyin Ma, Gongfan Fang, and Xinchao Wang. 2023. Llm-pruner: On the structural pruning of large language models. *Advances in neural information processing systems*, 36:21702–21720.

Xin Men, Mingyu Xu, Qingyu Zhang, Bingning Wang, Hongyu Lin, Yaojie Lu, Xianpei Han, and Weipeng Chen. 2024. Shortgpt: Layers in large language models are more redundant than you expect. *arXiv preprint arXiv:2403.03853*.

Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. 2016. Pointer sentinel mixture models. *Preprint*, arXiv:1609.07843.

Shervin Minaee, Tomas Mikolov, Narjes Nikzad, Meysam Chenaghlu, Richard Socher, Xavier Amatriain, and Jianfeng Gao. 2024. Large language models: A survey. *arXiv preprint arXiv:2402.06196*.

Humza Naveed, Asad Ullah Khan, Shi Qiu, Muhammad Saqib, Saeed Anwar, Muhammad Usman, Nick Barnes, and Ajmal Mian. 2023. A comprehensive overview of large language models. *arXiv preprint arXiv:2307.06435*.

Xuchen Pan, Yanxi Chen, Yaliang Li, Bolin Ding, and Jingren Zhou. 2024. Ee-tuning: An economical yet scalable solution for tuning early-exit large language models. *arXiv preprint arXiv:2402.00518*.

Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. 2019. Exploring the limits of transfer learning with a unified text-to-text transformer. *arXiv e-prints*.

Mohaimenul Azam Khan Raiaan, Md Saddam Hossain Mukta, Kaniz Fatema, Nur Mohammad Fahad, Sadman Sakib, Most Marufatul Jannat Mim, Jubaer Ahmad, Mohammed Eunus Ali, and Sami Azam. 2024. A review on large language models: Architectures, applications, taxonomies, open issues and challenges. *IEEE Access*.

Machel Reid, Edison Marrese-Taylor, and Yutaka Matsuo. 2021. Subformer: Exploring weight sharing for parameter efficiency in generative transformers. *arXiv preprint arXiv:2101.00234*.

Keisuke Sakaguchi, Ronan Le Bras, Chandra Bhagavatula, and Yejin Choi. 2021. Winogrande: An adversarial winograd schema challenge at scale. *Communications of the ACM*, 64(9):99–106.

Andrea Santilli, Silvio Severino, Emilian Postolache, Valentino Maiorca, Michele Mancusi, Riccardo Marin, and Emanuele Rodolà. 2023. Accelerating transformer inference for translation via parallel decoding. *arXiv preprint arXiv:2305.10427*.

Daria Soboleva, Faisal Al-Khateeb, Robert Myers, Jacob R Steeves, Joel Hestness, and Nolan Dey. 2023. Slimpajama: A 627b token cleaned and deduplicated version of redpajama. https://www.cerebras.net/blog/slimpajama-a-627b-token-cleaned-and-deduplicated-version-of-redpajama.

Sho Takase and Shun Kiyono. 2021. Lessons on parameter sharing across layers in transformers. *arXiv preprint arXiv:2104.06022*.

Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. 2023. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*.

Tycho FA van der Ouderaa, Markus Nagel, Mart Van Baalen, and Tijmen Blankevoort. 2024. The llm surgeon. In *The Twelfth International Conference on Learning Representations*.

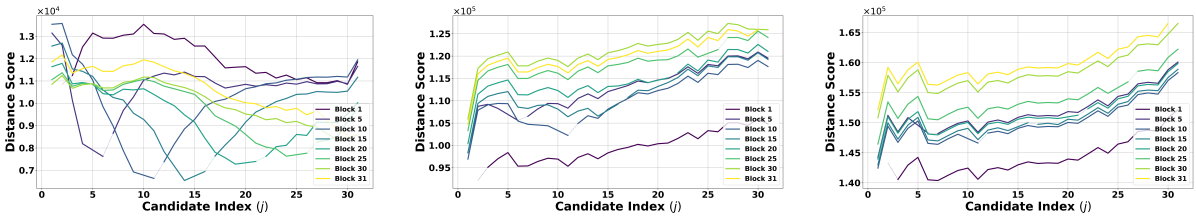
Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. *Advances in neural information processing systems*, 30.

Yifei Yang, Zouying Cao, and Hai Zhao. 2024. Laco: Large language model pruning via layer collapse. *arXiv preprint arXiv:2402.11187*.

Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali Farhadi, and Yejin Choi. 2019. Hellaswag: Can a machine really finish your sentence? *arXiv preprint arXiv:1905.07830*.

Peiyuan Zhang, Guangtao Zeng, Tianduo Wang, and Wei Lu. 2024. Tinyllama: An open-source small language model. *arXiv preprint arXiv:2401.02385*.

Susan Zhang, Stephen Roller, Naman Goyal, Mikel Artetxe, Moya Chen, Shuohui Chen, Christopher Dewan, Mona Diab, Xian Li, Xi Victoria Lin, et al. 2022. Opt: Open pre-trained transformer language models. *arXiv preprint arXiv:2205.01068*.



(a) Eq. (2) *with* high-rank pruning (b) Eq. (2) *without* high-rank pruning (c) Frobenius norm of $\mathbf{W}_i - \mathbf{W}_j$

Figure 4: Comparison of block distance score versus candidate block index j for different metrics. Dotted lines represent where candidate block index j is equal to pruning block index i, which is not a valid candidate. (a) Using the proposed metric in Eq. (2) with high-rank pruning, showing that closer blocks score lower (better), matching our intuition that weights close in the model have similar function. (b) Ablation of high-rank pruning, where there is no clear trend except that blocks closer to 0 are lower and those closer to 31 are higher. (c) Simple Frobenius norm, showing a similar lack of clear trend as in (b). *We found that using the score in (a) as the weight-sharing selection metric results in a much higher performing model compared to using the scores in (b) and (c).*

Appendix

A Additional Experimental Details

Our implementations were carried out using PyTorch, utilizing Fully Sharded Data Parallel (FSDP) and FP-16 mixed precision training for enhanced efficiency. The experiments were conducted on a setup comprising 4 NVIDIA A100 80GB GPUs and required $\approx$ 192 gpu hours per experiment. While we only report a single run per result, we evaluate on several models and several tasks. For model handling and training, we employed the Hugging Face Transformers library. We used a learning rate of 0.004 with a a cosine learning rate decay schedule, with a batch size of 2 per GPU and a total batch size of 480 achieved through gradient accumulation. The SlimPajamas dataset (Soboleva et al., 2023) train set was used, with 1B tokens dedicated to pruning experiments and 10B tokens for model extension experiments due to the faster processing speeds of the models. The LoRA rank utilized was 256. Compared to ShortGPT, our method incurs a 3.67% relative increase in total parameters for the main experiment setting of Table 1.

B Additional Method Analysis

In Figure 4, we show an alternative version of Figure 2 where the x-axis is candidate block index j instead of block index distance $(i - j)$. The purpose of this figure is to give an additional way to visualize the metrics which better highlights the issue in Figures 2(b) and 2(c) where all blocks i choose candidate block $j = 0$.

C Licenses of Datasets and Models

We used 1B tokens from the SlimPajama (Soboleva et al., 2023) pre-training dataset for post-prune recovery. For zero-shot performance evaluations, we used the ARC-e, ARC-c (Clark et al., 2018), PIQA (Bisk et al., 2020), WinoGrande (Sakaguchi et al., 2021), and HellaSwag (Zellers et al., 2019) zero-shot benchmarks, utilizing the LM Evaluation Harness (Gao et al., 2021). For perplexity performance evaluations, we used the validation MiniPile (Kaddour, 2023) subset of the Pile dataset (Gao et al., 2020). We confirmed that the data that was used does not contain any information that names or uniquely identifies individual people or offensive content by checking their distribution sources. All datasets use the English language. For the pruning experiments, we evaluated our method using LLaMA-2 7B (Touvron et al., 2023), OPT 1.3B and 6.7B (Zhang et al., 2022), and LLaMA-3 8B (AI@Meta, 2024), focusing on these due to their widespread adoption by the community. For the model extension experiments, we evaluated our method for model extension using TinyLLaMA (Zhang et al., 2024) due to its suitability for demonstrating the effectiveness of our approach in extending smaller models.

The licenses for the datasets and models used in this paper are as follows:

- **SlimPajama**: Apache License 2.0
- **ARC**: CC BY-SA 4.0
- **PIQA**: Academic Free License 3.0
- **HellaSwag**: MIT License